

Enterprise AI Runtime Optimization

Turin

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Description

Background

- The growing adoption of AI in the company requires common approaches that can be reused across projects instead of optimising only each solution independently.

Project Objective

Analyse and prototype improvements for the internal IT AI Platform and enterprise AI projects, focusing on three connected areas:

1. Reusable skills for prompt Optimization: Define modular and versioned AI skills that encapsulate instructions, constraints, examples and tool configurations. At runtime, only the skills relevant to the user request are selected and dynamically injected into the prompt, reducing unnecessary context while improving consistency, maintainability and reuse across projects..

2. Latency & efficiency: Reduce response time and token consumption in multi-agent workflows through better prompting, routing, context management and orchestration.

3. Data extraction: Define a reusable process to extract and structure information from heterogeneous documents such as PowerPoint, Excel and other unstructured sources.

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Additional info & features

Why is the project conducted?

Improve the efficiency, quality and scalability of enterprise AI solutions through reusable platform capabilities.

Who are the stakeholders?

Iveco IT AI Platform and CoE AI, with future AI project teams as potential users.

What are the main tasks?

- Map recurring activities and bottlenecks in enterprise AI projects.
- Establish baselines for latency, token consumption and extraction quality.
- Prototype generic extraction from heterogeneous documents.
- Skill-based Prompt Optimization: Instead of maintaining large, monolithic prompts, the solution will evaluate a modular prompting approach based on reusable skills. A lightweight routing layer identifies the intent and selects the most relevant skills from a governed catalogue. Only the required instructions and context are included in the model request, improving prompt efficiency and simplifying the evolution of multi-agent solutions.
- Prototype dynamic skill selection and prompt composition, injecting only the instructions and tools relevant to each request. Define a governed skill contract and measure its impact on latency, token usage, quality and reuse.
- Compare optimization approaches and provide recommendations for platform evolution.

Expected output

A demonstrator, a measured comparison of optimization approaches and reusable design recommendations for the IT AI Platform, including a prototype for dynamic skill selection and prompt composition. The assessment will compare baseline and skill-based executions in terms of latency, token consumption, output quality and maintainability. Course activities will use synthetic, open or non-confidential material.

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Mentoring

Given the project relevance, you will be supported and guided by:



Pasquale Morra
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IT COE AI&Data – AI Architecture

I contribute to the evolution of IVECO Group's IT AI Platform and enterprise AI architectures, focusing on reusable capabilities, multi-agent solutions, RAG, model integration, observability and governance. I support AI projects in adopting scalable patterns and common platform standards.



Marco Barbero
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IT COE AI&Data – Machine Learning

I oversee and monitor the implementation of artificial intelligence projects within the company. My main task is to directly supervise the development of machine learning solutions, but I also provide support for generative projects with the rest of the IVECO AI team.

Based on our existing commitments, we will be available for approximately one hour per week for mentoring activities, when required.


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